INCH-POUND
MIL-PRF-55342G
AMENDMENT 5
12 April 2002
SUPERSEDING
AMENDMENT 4
27 February 2001

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, FILM, CHIP,
NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, SPACE LEVEL,
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-PRF-55342G, dated 3 July 1997, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

TABLE II, row two, columns three and four, delete “Bonding pads” and “P”.

PAGE 4

TABLE III, column one, designator for 10 percent tolerance, delete and substitute:

“

1M00 to 8M20 inclusive
10M0 to 82M0 inclusive
100M to 820M inclusive
1N00 to 8N20 inclusive
10N0 to 82N0 inclusive
100N to 820N inclusive
1P00 to 8P20 inclusive
10P0

“

PAGE 10

3.9.1, add the following to end of paragraph, “and the change in resistance for group C tests shall not exceed the value specified in table I.”

PAGE 12

3.19.3, delete from paragraph title, “and P”.

PAGE 13

3.25, delete first sentence.

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PAGES 13 and 14

3.25.1 and 3.25.2, delete and substitute:

"3.25.1 Individual chip marking. Marking of individual chip resistors is not required but is permitted. Chip resistors may be marked using the following code systems. The marking requirement and code system used shall be specified on the order. If no code system is specified, the manufacturer may select the code. Abbreviated markings due to limited space takes precedence over PIN format.

"3.25.1.1 Option 1, four character part number code. Markings shall be the same four characters used to indicate resistance and tolerance in the PIN. See 1.2.5 and 1.2.6 and tables III and V.

Example: 30.1k ohms and 1 percent tolerance = 30E1
18.2 ohms and 1 percent tolerance = 18D2

"3.25.1.2 Option 2, four characters for 0.1 and 1.0 percent tolerances. The first three digits are significant figures and the last digit specifies the number of zeros to follow. When the value of resistance is less than 100 ohms, or when fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits of the group represent significant figures.

Example: 5.62 ohms = 5R62
14.7 ohms = 14R7
30.1 kohms = 3012

"3.25.1.3 Option 3, three characters for 2, 5, and 10 percent tolerances. The first two digits are significant figures and the last digit specifies the number of zeros to follow. When the value of resistance is less than 10 ohms, or when fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits for the group represents significant figures.

Example: 9.7 ohms = 9R7
330 kohms = 334
12 ohms = 120

"3.25.1.4 Option 4, four dots for 0.1 and 1.0 percent tolerances. Four color dots shall be present in the location and order as shown in the following example.

Example: 1st dot = 1st significant digit
2nd dot = 2nd significant digit
3rd dot = 3rd significant digit
4th dot = multiplier (Ref. MIL-STD-1285)

30.1 kohms = Orange Black Brown
Red

(Example as depicted in 3.25.1.2 of MIL-PRF-55342G).

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"3.25.1.5 Option 5, three color dots for 2, 5, and 10 percent tolerances. Three color dots shall be present in the location and order as shown in the following example.

Example: 1st dot = 1st significant digit
2nd dot = 2nd significant digit
3rd dot = multiplier (Ref. MIL-STD-1285)

330 kohms = Orange Orange Yellow

(Example as depicted in 3.25.1.1 of MIL-PRF-55342G).

"3.25.2 Option 6, three digit code for indicating value and tolerance. For 1 percent tolerance, the first two digits are numbers indicating standard decade values. The third character is a letter multiplier. For 2, 5, and 10 percent tolerances, the first character is a letter multiplier followed by a two digit number that indicates standard decade values as shown in the following examples.

THREE DIGIT CODE FOR 1 PERCENT TOLERANCE

(A = 10^0, B = 10^1, C = 10^2, D = 10^3, E = 10^4, R = 10^{-1}, S = 10^{-2})

Examples:	Resistance	Code
	1.00	01S
	12.1	09R
	165	22A
	4.32 MEG	62E

1 percent tolerance											
Symbol	Resistance	Symbol	Resistance	Symbol	Resistance	Symbol	Resistance	Symbol	Resistance	Symbol	Resistance
01	100	17	147	33	215	49	316	65	464	81	681
02	102	18	150	34	221	50	324	66	475	82	698
03	105	19	154	35	226	51	332	67	487	83	715
04	107	20	158	36	232	52	340	68	499	84	732
05	110	21	162	37	237	53	348	69	511	85	750
06	113	22	165	38	243	54	357	70	523	86	768
07	115	23	169	39	249	55	365	71	536	87	787
08	118	24	174	40	255	56	374	72	549	88	806
09	121	25	178	41	261	57	383	73	562	89	825
10	124	26	182	42	267	58	392	74	576	90	845
11	127	27	187	43	274	59	402	75	590	91	866
12	130	28	191	44	280	60	412	76	604	92	887
13	133	29	196	45	287	61	422	77	619	93	909
14	137	30	200	46	294	62	432	78	634	94	931
15	140	31	205	47	301	63	442	79	649	95	953
16	143	32	210	48	309	64	453	80	665	96	976

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THREE DIGIT CODE FOR 2, 5, and 10 PERCENT TOLERANCE

(A = 10^0, B = 10^1, C = 10^2, D = 10^3, E = 10^4, R = 10^{-1}, S = 10^{-2})

Examples:	Tolerance	Resistance	Code
	2-percent	1.00	S01
	5-percent	1.00	S25
	10-percent	1.00	S49
	5-percent	510	A42
	10-percent	820k	D60

2 percent		5 percent		10 percent	
Symbol	Resistance	Symbol	Resistance	Symbol	Resistance
01	100	25	100	49	100
02	110	26	110	50	120
03	120	27	120	51	150
04	130	28	130	52	180
05	150	29	150	53	220
06	160	30	160	54	270
07	180	31	180	55	330
08	200	32	200	56	390
09	220	33	220	57	470
10	240	34	240	58	560
11	270	35	270	59	680
12	300	36	300	60	820
13	330	37	330		
14	360	38	360		
15	390	39	390		
16	430	40	430		
17	470	41	470		
18	510	42	510		
19	560	43	560		
20	620	44	620		
21	680	45	680		
22	750	46	750		
23	820	47	820		
24	910	48	910		

"

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3.25.5, first sentence, between “characteristics” and “or”, add “with comparable power ratings” and at end of first sentence, delete ”(see table VII)” and substitute “in accordance with table VII”

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TABLE VIII, group IV, third column, add “or 4.8.11.2”.

TABLE VIII, footnote 3/, delete “and P”.

PAGE 21

4.6.3.2.4, last sentence, delete "for the sample quality" and substitute "for the same quality".

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TABLE X, delete and substitute:

"TABLE X. Sampling plans for subgroup 3 and subgroup 4 and ppm assessment.

Lot size	Sample size for subgroup 3	Sample size for subgroup 4	Sample size for ppm assessment
1 to 5	100%	100%	100%
6 to 13	100%	5	100%
14 to 125	13	5	100%
126 to 150	13	5	125
151 to 280	20	5	125
281 to 500	29	5	125
501 to 1,200	34	5	125
1,201 to 3,200	42	5	125
3,201 to 10,000	50	8	125
10,001 to 35,000	60	13	294
35,001 to 150,000	74	20	294
150,001 to 500,000	90	20	345
500,001 and over	102	20	435

"

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TABLE XI, footnote 5/, delete “and P”.

PAGE 25

4.7.1.1, title, delete “Subgroup 1” and substitute “Monthly (subgroup 1)”.

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4.7.1.2, title, delete “Subgroup 2” and substitute “Monthly (subgroup 2)”.

4.7.2.2, end of sentence, delete “in the order shown”.

4.7.2.3, first sentence, delete “and P”.

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4.8.1.3.a, first and second sentence, delete “P and”.

4.8.1.3.a(1), between “D” and “T” add “G”

4.8.1.3.b, first sentence, delete “and P”.

4.8.1.3.c, between “C” and “T” add “G” and delete “in accordance with MMM-A-1931” and substitute “approved by the qualifying activity”

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* 4.8.3.b(1), first line, delete “Qualification and group A tests:” and substitute “Qualification test:”

TABLE XIV, delete and substitute:

"TABLE XIV. Resistance temperature characteristic.

Sequence	Temperature 1/
	°C
1	2/ 25 ±3
2	-55 ±3
3	2/ 25 ±3
4	125 ±3

1/ At the option of the manufacturer, the reverse sequence may be as follows:
1 - 1/ 25 ±3
2 - 125 ±3
3 - 1/ 25 ±3
4 - -55 ±3

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures."

4.8.13.1, add new third sentence, "For style RM0402, the force applied shall be 1 kilogram."

4.8.13.1, original third sentence between "RM0505 and RM1005", add "RM0603"

4.8.13.2, add new third sentence, "For style RM0402, the force applied shall be 0.5 kilogram."

4.8.13.2, original third sentence between "RM0505 and RM1005", add "RM0603"

* 4.8.14.a, add “and mounted”

6.2.c, delete and substitute:

“Packaging requirements (e.g., Electrostatic discharge (ESD) sensitivity) (see 5.1).

6.6, add the following new sentence:

“Direct shipments to the Government is controlled by MIL-DTL-39032 which specifies a preventive packaging procedure.”

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A.3.2, delete in its entirety and substitute:

"A.3.2 Sample. A sample consisting of 162 sample units, 54 of the lowest, 54 of the highest, and 54 of the critical or nearest to critical value (see table XV), and in each style and characteristic, termination material, and resistance tolerance for which qualification is sought, shall be submitted and subjected to the inspection of table VIII. If no critical value is specified, the sample will be equally divided between the highest and lowest resistance values. Ten sample units of any resistance value or characteristic shall be submitted and subjected to the test of group V. Ten sample units of any resistance value shall be submitted to tests of group VI, when applicable. Ten sample units shall be submitted to the test in group VII. If qualification to a closer resistance tolerance than submitted above is desired, 30 additional sample units of the closer resistance tolerance shall be subjected to the following tests (see table XV). After qualification has been granted, no changes shall be made in materials, design, or construction without prior notification to the qualifying activity."

A.3.1.1, delete and substitute "A.3.2.1".

Add the following:

"A.3.3 Outgassing (space level only). Manufacturers shall validate to the qualifying activity the outgassing requirements as specified in 3.24."

TABLE XVI, add the two new styles,

"TABLE XVI. Critical resistance values for qualification ranges. 1/

Style	Slash sheet	Critical resistance value 2/ 3/							
		Characteristic							
		E		H		K		M	
		Ceramic board	Fiber board	Ceramic board	Fiber board	Ceramic board	Fiber board	Ceramic board	Fiber board
RM0402	/11	15.4	15.4	15.4	15.4	15.4	15.4	15.4	15.4
RM0603	/12	35.7	35.7	35.7	35.7	35.7	35.7	35.7	35.7

"

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TABLE XVII, Row 2, delete and substitute:

Style	Will qualify style 1/
RM0402	RM0402
RM0502	RM0502, RM0402
RM0603	RM0603, RM0502, RM0402
RM0505	RM0505
RM0705	RM0705, RM0603, RM0505
RM1005	RM1005, RM0705, RM0505
RM1010	RM1010, RM1005, RM0705, RM0505
RM1505	RM1505, RM1005, RM0705, RM0505
RM2208	RM2208
RM1206	RM1206, RM1505, RM1010, RM1005, RM0705, RM0505
RM2010	RM2208, RM2010
RM2512	RM2010, RM2208, RM2512

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TABLE XVII, under termination material, delete all termination “P” and revise for “G” and “B” as follows:

"

Termination	Will qualify termination	May qualify termination with minor additional testing 2/
G B	G B	B, W G, W, C, U 3/

"

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FIGURE 7, delete paragraph references and substitute:

Delete	Substitute
3.23.1.2	B.3.1.2
3.23.1.3	B.3.1.3
3.23.1.4	B.3.1.4.a
3.23.1.4	B.3.1.4.b
3.23.1.5	B.3.1.5
3.23.1.6	B.3.1.6
3.23.1.7	B.3.1.7

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